

# Catálogo de Átomos de Confusão em SystemVerilog

1.

Lambda - com smell

```
module mux (  
    input logic [1:0] selector,  
    input logic enable,  
    output logic [7:0] out  
);  
    logic [7:0] decoded;  
  
    always_comb begin  
        case (selector)  
            0: decoded = 8'd10;  
            01: decoded = 8'd20;  
            2: decoded = 8'd30;  
            10: decoded = 8'd40;  
            default: decoded = 8'd100;  
        endcase  
  
        if (enable)  
            out = decoded;  
        else  
            out = 8'd0;  
        end  
    end  
endmodule
```

**Gabarito:**

**selector = 2'b10**

**enable = 1 -> out = 30**



Omega - sem smell

```
module mux (  
    input logic [1:0] selector,  
    input logic enable,  
    output logic [7:0] out  
);  
    logic [7:0] decoded;  
  
    always_comb begin  
        case (selector)  
            2'b00: decoded = 8'd10;  
            2'b01: decoded = 8'd20;  
            2'b10: decoded = 8'd30;  
            2'b11: decoded = 8'd40;  
            default: decoded = 8'd100;  
        endcase  
  
        if (enable)  
            out = decoded;  
        else  
            out = 8'd0;  
        end  
    end  
endmodule
```

**Gabarito:**

**selector = 2'b10**

**enable = 1 -> out = 30**

## 2.

## Omega - sem smell

```
out0 = 0000000000000000000000000000000011  
out1 = 0000000000000000000000000000000000
```

```
out0 = 0000000000000000000000000000000011  
out1 = 0000000000000000000000000000000000
```

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4.

Lambda - com smell

```
module example (  
    input logic enable,  
    output int out  
);  
  
    int result;  
    byte in = 5;  
  
    always_comb begin  
        result = in + 1'b1;  
  
        if (enable)  
            out = result;  
        else  
            out = in;  
    end  
endmodule
```

**Gabarito:**  
enable = 1 -> out = 6

Omega - sem smell

```
module example (  
    input logic enable,  
    output int out  
);  
  
    int result;  
    byte in = 5;  
  
    always_comb begin  
        result = in + 1;  
  
        if (enable)  
            out = result;  
        else  
            out = in;  
    end  
endmodule
```

**Gabarito:**  
enable = 1 -> out = 6

# Catálogo de Átomos de Confusão em SystemVerilog

## 5.

### Lambda - com smell

```
module example;  
  
  int lo, med, hi;  
  bit result;  
  int selected;  
  
  initial begin  
    lo = 20;  
    med = 164;  
    hi = 224;  
  
    result = lo < med < hi;  
  
    if (result)  
      selected = med;  
    else  
      selected = hi;  
  
  end  
  
endmodule
```

**Gabarito:**  
**result = 1**  
**selected = 164**

### Omega - sem smell

```
module example;  
  
  int lo, med, hi;  
  bit result;  
  int selected;  
  
  initial begin  
    lo = 20;  
    med = 164;  
    hi = 224;  
  
    result = (lo < med) && (med < hi);  
  
    if (result)  
      selected = med;  
    else  
      selected = hi;  
  
  end  
  
endmodule
```

**Gabarito:**  
**result = 1**  
**selected = 164**

